Voice Creator – Prototype Development Plan

Prototype Plan Report

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1. Introduction

This report outlines the prototype development plan for the project **“Voice Creator”**, a modular text-to-speech (TTS) application. The goal of the prototype is to validate the feasibility of customizable voice generation and demonstrate the core functionality before full-scale development.

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2. Overview of the Proposed Application

Voice Creator is designed to allow users to convert text into speech with advanced customization options. Unlike traditional TTS systems, this application focuses on modular control, enabling users to adjust multiple voice parameters and create personalized audio outputs.

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3. Key Features of the Prototype

The prototype will focus on the core functionalities required to validate the idea:

- **Text Input Module:** Users can enter or paste text for conversion
- **Basic Text-to-Speech Generation:** Convert text into audible speech
- **Voice Customization Controls:** Adjust parameters such as pitch, speed, and tone
- **Real-Time Preview:** Instantly listen to generated speech
- **Simple User Interface:** Minimal interface to test usability and interaction

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4. Role of AI Studio in Prototype Development

AI Studio will be used as the primary tool for building and testing the prototype’s core functionality.

- Enables **AI-based speech generation** without building models from scratch
- Allows experimentation with **different voice styles and tones**
- Helps simulate **modular voice control using parameter adjustments**
- Provides quick iteration and testing of **voice outputs and quality**

AI Studio will significantly reduce development time and allow focus on validating the concept rather than infrastructure.

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5. Prototype Validation Strategy

The prototype will be used to:

- Test whether customizable voice output is practical and effective
- Evaluate the quality and naturalness of generated speech
- Gather feedback on usability and feature requirements
- Identify technical limitations and improvements

This validation ensures that the idea is feasible before investing in full development.

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6. Implementation Using Android Studio

After successful validation, the application will be developed using **Android Studio**.

Planned implementation steps include:

- Designing a **user-friendly interface** using modern UI frameworks (e.g., Jetpack Compose)
- Integrating **text-to-speech APIs or AI services** tested in the prototype
- Implementing **interactive controls** (sliders, buttons) for voice customization
- Adding features like **saving voice profiles and exporting audio files**
- Managing application state and performance optimization

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7. Benefits of the Development Approach

- Reduces development risk through early validation

- Ensures a **user-centered design**
- Improves efficiency by separating prototyping and implementation phases
- Enables faster iteration and refinement

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8. Conclusion

The prototype development plan for Voice Creator focuses on validating the core concept of modular voice customization using AI Studio. Once validated, Android Studio will be used to build a complete, user-friendly application. This structured approach ensures a reliable, scalable, and effective final product.